LUCIA WEINER

(240) 205-6777 www.linkedin.com/in/lucia-weiner/ luciagweiner@gmail.com

EDUCATION

University of Florida (UF), ***Herbert Wertheim College of Engineering*****,** Gainesville, FL **May 2028**
Bachelor of Science in Industrial and Systems Engineering, Minor in Computer Science GPA 3.84/4.00

Relevant Coursework: Matrix and Numerical Methods for Systems Engineering, Data Analysis for Industrial Applications, Elements of Electrical Engineering, Data Structures and Algorithms, Computational Linear Algebra, Discrete Mathematics

EXPERIENCE

ExxonMobil Global Trading - Houston, TX **January 2026 - August 2026**
Crude Trading - Software Enablement Intern

- Implemented an AI agent that parses Zoom messages and Excel sheets to automatically generate entries for international crude oil export deal trackers, automating data entry for the US Gulf Coast and East Canada regions
- Designed a scalable validation pipeline that normalizes data source formats and validates fields before insertion, reducing data entry errors and safeguarding accuracy for commercial trading decisions
- Reduced manual workload time by 89%, enabling review and approval of AI-suggested export deal entries rather than hands-on processing of Zoom transcripts, emails, and Excel files

Crude Trading - Analytics Intern

- Built a Databricks Python pipeline to ingest weekly crude oil pipeline flow data from the Genscape API, implemented data transformations, and loaded raw and curated datasets into Snowflake to support analytics and trade decision-making
- Automated forecasting logic to predict Cushing inventory levels, eliminating manual spreadsheet processes and reducing monthly data manipulation time by 10 hours
- Designed and built an interactive frontend UI in React and TypeScript to visualize the processed pipeline data and predicted supply/demand balance, enabling faster and more informed trading decisions

UF Human Performance & Neuro-Engineering Lab - Gainesville, FL **June 2025 - January 2026**
Research Assistant

- Co-authored a research paper on a wearable sensor framework for real-time health and human performance monitoring in operational environments
- Published in IEEEXplore and selected as a special session paper at the 2025 First International Conference on Intelligent Computing and Systems at the Edge (ICEdge)
- Conducted a literature review of doctoral research in biomechanics and human-robot interaction, focused on exploring design trade-offs between assistive performance and wearer comfort in exoskeleton systems

University of Texas at Austin Computer Science Department - Austin, TX **June - July 2023 & 2025**
Technical Program Intern

- Collaborated with a team of five to teach a C++ robotics and Python machine learning curriculum, achieving an 80% robot assembly rate among 300+ students
- Orchestrated interactive C++ robotics simulations using the Arduino IDE and led hands-on hardware workshops
- Taught introductory machine learning lessons on Perceptron and Q-learning algorithms, guiding students through hands on implementations in Python

PROJECTS

AI Allergen Cooking Assistant - Gainesville, FL **June 2025 - January 2026**

- Built a retrieval-augmented generation system that answers cooking questions using uploaded PDF cookbooks
- Implemented PDF chunking, embedding generation, and vector database querying to improve context-aware responses

Financial Trading Strategy Project - Gainesville, FL **January 2025 - May 2025**

- Served as software engineering lead in a team of four engineers, directing the implementation of a custom trading algorithm that utilized Exponential Moving Averages (EMAs) and Bollinger Bands to produce buy/sell signals
- Conducted back testing on our EMA-based trading model, leveraging historical data to evaluate profitability, achieving a 40% profit that outperformed a buy-and-hold S&P 500 strategy by ~181% in 2024

LEADERSHIP

Theta Tau Professional Engineering Fraternity - Gainesville, FL **January 2025 - Present**
Professional Development Chair

- Led initiatives to enhance career readiness by facilitating resume reviews, and personalized interview preparation

ADDITIONAL INFORMATION

Technical skills: Java, Python, C/C++, HTML/CSS, JavaScript, TypeScript, React, MATLAB, Arduino, Microsoft Office Suite
Awards: President's Honor Roll, Dean's List, UF Distinguished Scholar Merit Award